

Social Media and Mental Health Balance

A Statistical and Data Analysis Project Report

Dataset Source: Kaggle | Prepared by: Hashmat Ullah Sultani

1. HOW DATA WAS COLLECTED

The dataset used in this project was sourced from **Kaggle**, a widely used platform for data science and statistics projects. It is specifically suited for Exploratory Data Analysis (EDA). The data was originally collected using online survey tools such as **Google Forms** and **SurveyMonkey**.

2. SOURCE AND ORIGIN

- **Primary Source:** This dataset was originally developed as part of a Statistics and Data Science course project at the University of Liberal Arts Bangladesh (ULAB).
- **Dataset Type:** Publicly available on Kaggle; commonly used for educational and research purposes in data analysis and applied statistics.

3. DESCRIPTIVE STATISTICS

The following section provides a detailed explanation of the key statistical measures derived from the dataset. Each variable is analyzed in terms of its mean, range, and practical significance.

3.1 Happiness Index

The mean **Happiness Index** score is **8.38**, indicating overall high self-reported happiness among participants. The lowest recorded score is 4, while the 75th percentile reaches 10, showing that the distribution is significantly skewed toward the higher end of the scale. This suggests that the majority of participants report a positive sense of well-being.

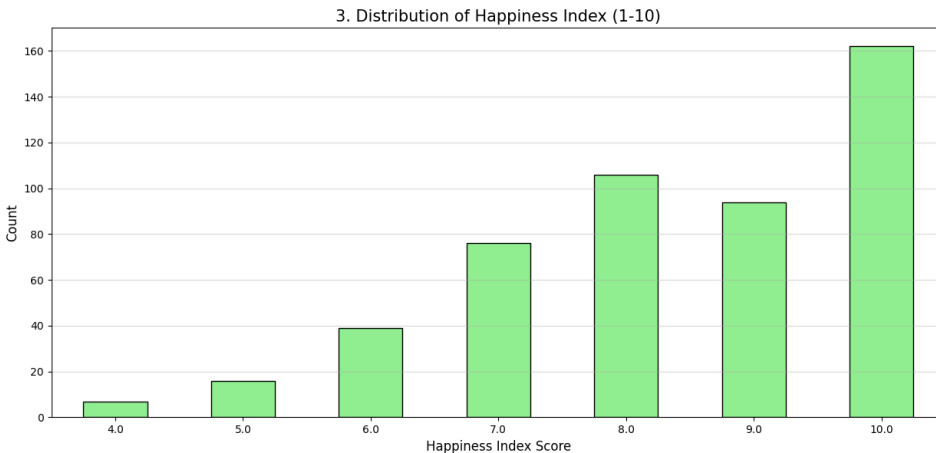


Figure 1: Distribution of Happiness Index

3.2 Daily Screen Time (hrs.)

Daily screen time ranges from **1 hour to nearly 11 hours**, with an average of **5.53 hours** per day. Measuring screen time is critical for understanding digital balance, as excessive screen time is consistently associated with negative mental health outcomes such as reduced happiness, disrupted sleep, and increased stress.

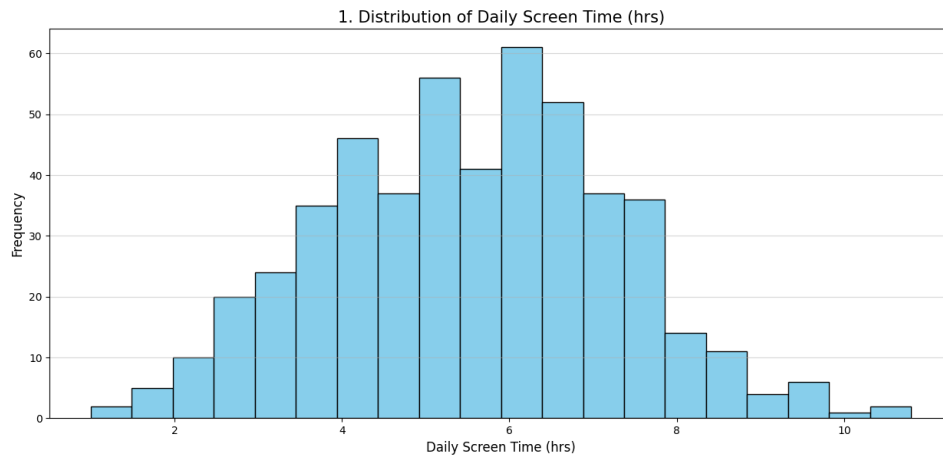


Figure 2: Distribution of Daily Screen Time

3.3 Stress Level (1-10)

Stress levels range from **2 to 10**, with a mean of **6.62**. This indicates that more than half of the participants fall into the high-stress category. Notably, while most participants report a high Happiness Index, they simultaneously experience significant stress — a pattern that suggests happiness and stress can coexist, likely moderated by other lifestyle factors.

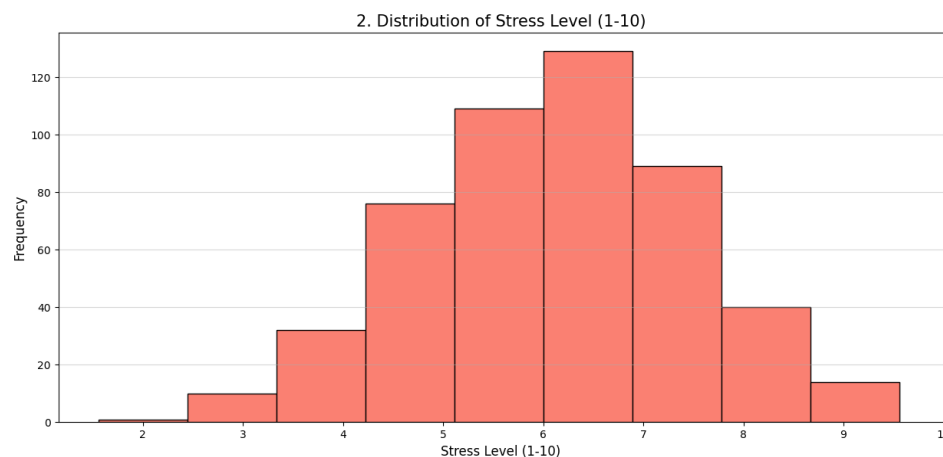


Figure 3: Distribution of Stress Levels

3.4 Sleep Quality (1-10)

The average **Sleep Quality** score is **6.30**, which is slightly above the midpoint of the scale. While generally acceptable, there is clear room for improvement. Sleep quality is directly linked to both screen time and stress levels — particularly for individuals who use social media or electronic devices immediately before bed, as exposure to screens before sleep is well documented to reduce sleep quality and duration.



Figure 4: Distribution of Sleep Quality

3.5 Exercise Frequency (per week)

Physical activity is a key factor in maintaining strong mental health. However, the average exercise frequency among participants is only **2.45 times per week**. This relatively low figure suggests that many participants do not exercise regularly, which may negatively affect both their physical health and psychological well-being over time.

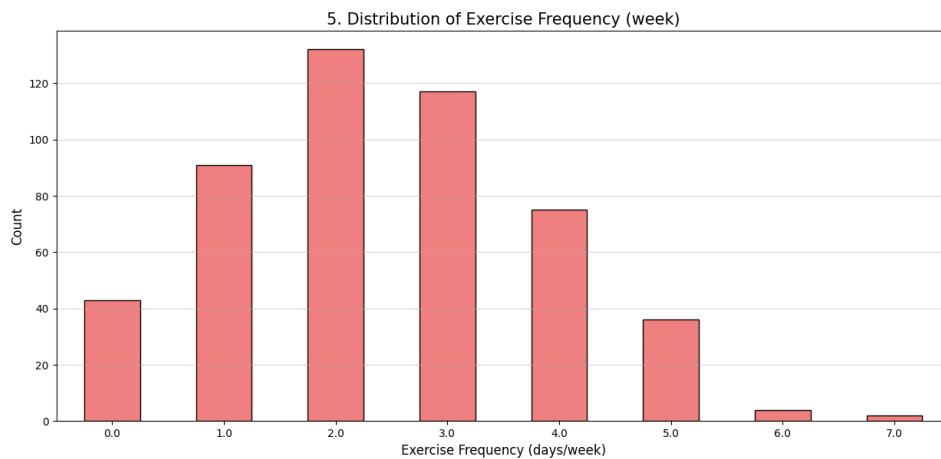


Figure 5: Distribution of Exercise Frequency

3.6 Categorical Variable Overview

The gender distribution of the dataset consists of **248 males**, **229 females**, and **23 individuals** who identified as "Other." Regarding social media platform usage, **TikTok** is the most frequently used platform, followed by **X (Twitter)** and **LinkedIn**.

Summary

The descriptive statistics for this dataset indicate a population that reports **high happiness** but also experiences **above-average stress levels** and significant daily social media exposure. These patterns suggest a complex relationship between digital habits and mental well-being that warrants deeper statistical analysis.

4. CORRELATION ANALYSIS

The Pearson correlation coefficient between **Daily Screen Time** and **Happiness Index** is $r = -0.7052$. This indicates a **strong negative relationship** between the two variables: as daily screen time increases, self-reported happiness tends to decrease. The negative sign reflects an **inverse relationship**, and the magnitude of the coefficient (close to -1) confirms that this association is substantial, providing strong statistical evidence that excessive digital usage is linked to lower levels of happiness.

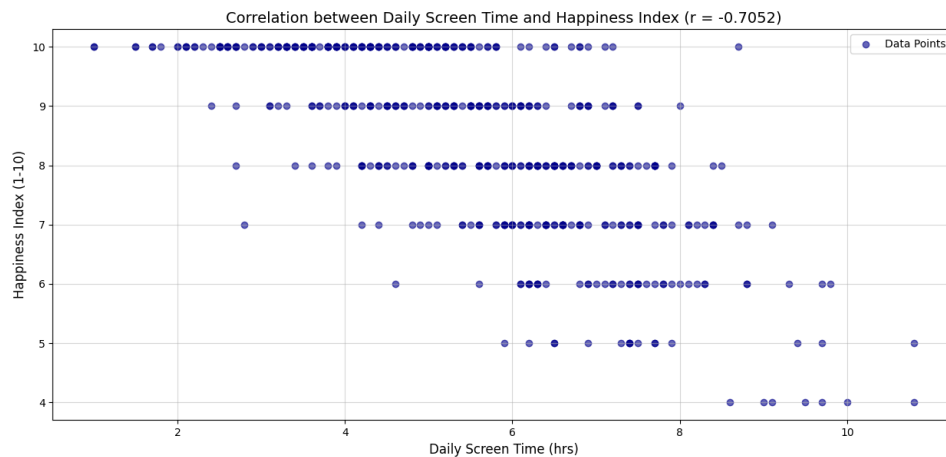


Figure 6: Correlation — Daily Screen Time vs. Happiness Index

5. REGRESSION ANALYSIS

A simple linear regression model was applied to quantify the predictive relationship between daily screen time and happiness. The model takes the standard form: **Happiness Index = $m \times$ Screen Time + c**

- **Slope (m):** The regression coefficient for screen time is -0.62 , meaning that for every additional hour of screen time, the predicted Happiness Index decreases by 0.62 points.
- **Intercept (c):** The intercept value is 11.80 , representing the predicted happiness score when screen time equals zero.
- **Full Equation:** Happiness Index = $-0.62 \times$ Daily Screen Time + 11.80
- **R^2 (Coefficient of Determination):** $R^2 = 0.50$, meaning that approximately 50% of the variance in happiness scores is explained by screen time alone.

These results demonstrate a **measurable and practically significant** impact of digital screen habits on well-being. The model indicates that happiness levels are meaningfully influenced by daily digital behavior, with higher screen time associated with a predictable and consistent decline in happiness scores.



Figure 7: Regression Line — Screen Time vs. Happiness Index

6. HYPOTHESIS TESTING: GENDER AND HAPPINESS

6.1 Hypothesis Setup

- **Null Hypothesis (H_0):** The mean Happiness Index of males is equal to the mean Happiness Index of females.
- **Alternative Hypothesis (H_1):** The mean Happiness Index of males is significantly different from that of females.
- **Significance Level:** $\alpha = 0.05$

6.2 Test Results and Conclusion

A **permutation test** was conducted to determine whether gender significantly influences happiness levels. The test produced a **p-value of 0.9840**. Since this value is substantially greater than the standard significance level of $\alpha = 0.05$, we **fail to reject the null hypothesis**. The permutation test confirms that there is no statistically significant difference between male and female happiness scores in this dataset. The data therefore suggests that happiness levels are consistent across gender groups.

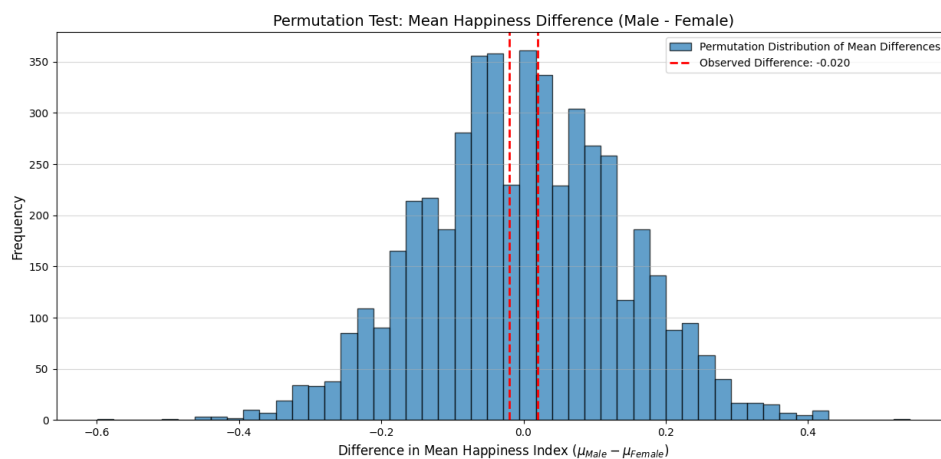


Figure 8: Happiness Distribution by Gender

7. CONCLUSION

This project analyzed the relationship between social media usage and mental health indicators using statistical methods applied to a real-world dataset of 500 participants. The key findings are as follows:

- **Screen Time & Happiness:** A strong negative correlation ($r = -0.71$) was found between daily screen time and happiness, with each additional hour of screen time associated with a 0.62-point decrease in happiness score.
- **Regression Model:** The regression model ($R^2 = 0.50$) confirms that screen time is a significant predictor of happiness, explaining half of its variance.
- **Gender & Happiness:** No statistically significant difference in happiness levels was found between gender groups ($p = 0.984$).
- **Lifestyle Indicators:** Participants report above-average stress (mean = 6.62) and below-average exercise frequency (mean = 2.45/week), suggesting that digital habits interact with broader lifestyle patterns.

Overall, this analysis demonstrates the value of data-driven methods in understanding the psychological impact of social media. Balanced digital consumption and healthy lifestyle habits appear to be important factors in maintaining mental well-being among young populations.